TVS MOBILITY

TVS SENSING SOLUTIONS

CORPORATE PRESENTATION

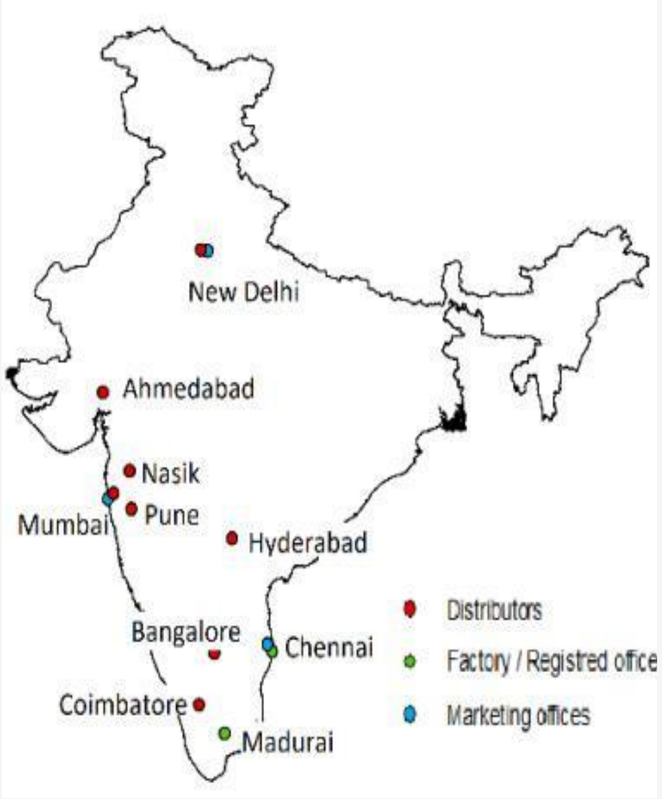
TVS SENSING SOLUTIONS

www.tvsss.co.in

TVS MOBILITY

AS TECHNOLOGY RAPIDLY EVOLVES, SO DO WE.

TVS SENSING SOLUTIONS



4 Sales offices and 10 Distributors



Fiber Optic Sensing Solutions (FOSS) Technology development center: Mumbai	
Category	**Target Market segment**
PIDS-FOPIDS, DAS-H	Home land security
	Oil & Gas, Defence (Border Intrusion Detection), Critical asset monitoring
Measurement-Temperature, Strain	Energy, Medical, Infrastructure and process industry
Others – Radar, Satellite	Defence Homeland security

Certification

Our certifications are a testament to our excellence and leadership in the auto component industry. They validate our commitment to quality, precision, and innovation, setting us apart as a top-tier provider in the market.



ISO 9001
IATF 16949



ISO 14001
OSHAS 45001





TVS MOBILITY

Product Certification

UL, VDE, and ENEC approved for switch manufacturing



DNV·GL

EU-TYPE EXAMINATION CERTIFICATE

[2] EQUIPMENT OR PROTECTIVE SYSTEM INTENDED FOR USE IN POTENTIALLY EXPLOSIVE ATMOSPHERES DIRECTIVE 2014/34/EU

[3] EU-Type Examination Certificate Number: **Presafe 19 ATEX 14300X** Issue 0

[4] Product: **Reed Sensor**

[5] Manufacturer: **TVS Sensing Solutions Private Limited**

[6] Address: **Madurai - Melur Road, Vellaripatti MADURAI. TAMIL NADU. India - 625 122**

[7] This product and any acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

[8] DNV GL Presafe AS, notified body number 2460, in accordance with Article 17 of Directive 2014/34/EU of the European Parliament and of the Council, dated 26 February 2014, certifies that this product has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of products intended for use in potentially explosive atmospheres given in Annex II to the Directive.

The examination and test results are recorded in confidential reports listed in section 16.

[9] Compliance with the Essential Health and Safety Requirements has been assured by compliance with:
EN 60079-0:2012/A11:2013 and EN 60079-18: 2015

[10] If the sign "X" is placed after the certificate number, it indicates that the product is subject to the Specific Conditions of Use specified in the schedule to this certificate.

[11] This EU - TYPE EXAMINATION CERTIFICATE relates only to the design and construction of the specified product. Further requirements of the Directive apply to the manufacturing process and supply of this product. These are not covered by this certificate.

[12] The marking of the product shall include the following:

Ex II 2 G Ex mb IIC T5/T6 Gb

Date of issue: 2019-05-23

Aste Kaastad
For DNV GL Presafe AS

PROD 021

This certificate may only be reproduced in its entirety and without any change, schedule included.

DNV GL Presafe AS, Veritasveien 3, 1363 Høvik, Norway, Tel +47 67 57 99 00, www.dnvgl.com

ATEX



DNV

[1]

PRODUCTION QUALITY ASSURANCE NOTIFICATION

[2] Equipment and Protective Systems Intended for use in Potentially Explosive Atmospheres Directive 2014/34/EU

[3] Notification Number: **Presafe 19 ATEX 14537Q** Issue 1

[4] Equipment, Protective System or components as listed: **Reed Sensor**
Type of protection: Ex m

(The EU-Type Examination Certificates based on this notification are listed by the notified body)

[5] Manufacturer or Authorized representative in the Community: **TVS Sensing Solutions Private Limited Madurai-Melur road, Vellaripatti, Madurai- 625 122, India**

[6] Manufacturing Locations: **TVS Sensing Solutions Private Limited Madurai-Melur road, Vellaripatti, Madurai- 625 122, India**

[7] DNV Product Assurance AS, notified body number 2460 for Annex IV, Module D in accordance with Article 17 of the Council Directive 2014/34/EU of 26 February 2014 notifies to the applicant that the manufacturer has a production quality system which complies with Annex IV, Module D of the Directive

This quality system in compliance with Annex IV, Module D of the Directive also meets the requirements of Annex VII, Module E.

[8] This notification is based on audit report listed on page 2

This notification can be withdrawn if the manufacturer no longer satisfies the requirements of Annex IV, Module D.

Results of periodical assessments of the quality system are a part of this notification.

[9] This notification is valid until **2025-03-21** and can be withdrawn if the manufacturer does not satisfy quality assurance surveillance of the production process

[10] According to Article 16 (3) of the Directive 2014/34/EU the CE mark shall be followed by the identification number 2460 identifying the Notified Body involved in the production control phase.

Date of issue: 2022-03-28

Ståle Sandstad
For DNV Product Assurance AS

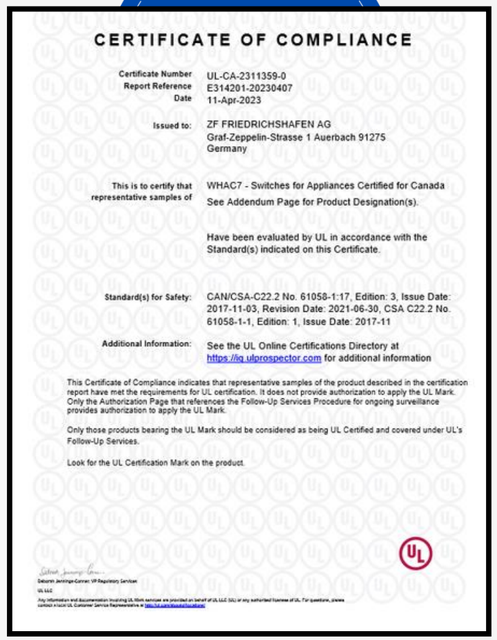
PROD 021

This certificate may only be reproduced in its entirety and without any change, schedule included.

DNV Product Assurance AS, Veritasveien 3, 1363 Høvik, Norway, Tel +47 67 57 99 00, www.dnv.com

ICP-4-6-II-ATEX-AL rev 1

ATEX



CERTIFICATE OF COMPLIANCE

Certificate Number: UL-CA-2311359-0
Report Reference: E314201-20230407
Date: 11-Apr-2023

Issued to: ZF FRIEDRICHSHAFEN AG
Graf-Zeppelin-Strasse 1 Auerbach 91275
Germany

This is to certify that representative samples of: WHAC7 - Switches for Appliances Certified for Canada
See Addendum Page for Product Designation(s).

Have been evaluated by UL in accordance with the Standard(s) indicated on this Certificate.

Standard(s) for Safety: CAN/CSA-C22.2 No. 61058-1:17, Edition: 3, Issue Date: 2017-11-03, Revision Date: 2021-06-30, CSA C22.2 No. 61058-1-1, Edition: 1, Issue Date: 2017-11

Additional Information: See the UL Online Certifications Directory at https://iq.ulprospector.com for additional information

This Certificate of Compliance indicates that representative samples of the product described in the certification report have met the requirements for UL certification. It does not provide authorization to apply the UL Mark. Only the Authorization Page that references the Follow-Up Services Procedure for ongoing surveillance provides authorization to apply the UL Mark.

Only those products bearing the UL Mark should be considered as being UL Certified and covered under UL's Follow-Up Services.

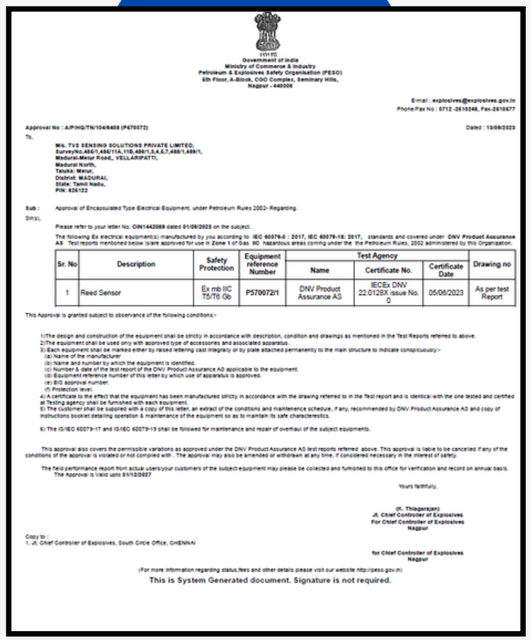
Look for the UL Certification Mark on the product.

Deborah Jennings-Conner, VP Regulatory Services

UL LLC

Any information and documentation involving UL Mark services are provided on behalf of UL LLC (UL) or any authorized licensee of UL. For questions, please contact a local UL Customer Service Representative at http://ul.com/aboutul/locations

UL



Government of India
Ministry of Commerce & Industry
Petroleum & Explosives Safety Organisation (PESO)
6th Floor, A-Block, CGO Complex, Seminary Hills, Nagpur - 440008

E-mail : explosives@explosives.gov.in
Phone/Fax No : 0712-2610348, Fax-2610077

Approval No : A/P/HQ/TN/104/8408 (P670072) Dated : 15/09/2023

To,
M/s. TVS SENSING SOLUTIONS PRIVATE LIMITED,
Survey/no.45/1,45/2/1A,1/2B,48/1/1,2,4,6,7,48/1,48/6/1,
Madurai-Melur Road., VELLARIPATTI,
Madurai North,
Taluka: Melur,
District: MADURAI,
State: Tamil Nadu,
PIN: 625122

Sub : Approval of Encapsulated Type Electrical Equipment, under Petroleum Rules 2002- Regarding.

Sir(s),

Please refer to your letter No. GN/1442089 dated 01/09/2023 on the subject.

The following Ex electrical equipment(s) manufactured by you according to IEC 60079-0 : 2017, IEC 60079-18: 2017, standards and covered under DNV Product Assurance AS Test reports mentioned below is/are approved for use in Zone 1 of Gas IIC hazardous areas coming under the the Petroleum Rules, 2002 administered by this Organization.

Sr. No	Description	Safety Protection	Equipment reference Number	Test Agency			Drawing no
				Name	Certificate No.	Certificate Date	
1	Reed Sensor	Ex mb IIC T5/T6 Gb	P570672/1	DNV Product Assurance AS	IECEx DNV 22.0128X Issue No. 0	05/06/2023	As per test Report

This Approval is granted subject to observance of the following conditions:-

1)The design and construction of the equipment shall be strictly in accordance with description, condition and drawings as mentioned in the Test Reports referred to above.
2)The equipment shall be used only with approved type of accessories and associated apparatus.
3) Each equipment shall be marked either by raised lettering cast integrally or by plate attached permanently to the main structure to indicate conspicuously:-
(a) Name of the manufacturer
(b) Name and number by which the equipment is identified.
(c) Number & date of the test report of the DNV Product Assurance AS applicable to the equipment.
(d) Equipment reference number of this letter by which use of apparatus is approved.
(e) BIS approval number.
(f) Protection level.
4) A certificate to the effect that the equipment has been manufactured strictly in accordance with the drawing referred to in the Test report and is identical with the one tested and certified at Testing agency shall be furnished with each equipment.
5) The customer shall be supplied with a copy of this letter, an extract of the conditions and maintenance schedule, if any, recommended by DNV Product Assurance AS and copy of instructions booklet detailing operation & maintenance of the equipment so as to maintain its safe characteristics.
6) The IS/IEC 60079-17 and IS/IEC 60079-19 shall be followed for maintenance and repair of overhaul of the subject equipments.

This approval also covers the permissible variations as approved under the DNV Product Assurance AS test reports referred above. This approval is liable to be cancelled if any of the conditions of the approval is violated or not complied with. The approval may also be amended or withdrawn at any time, if considered necessary in the interest of safety.

The field performance report from actual users/your customers of the subject equipment may please be collected and furnished to this office for verification and record on annual basis.
The Approval is Valid upto 01/10/2027

Yours faithfully,

(R. Thiagarajan)
JS, Chief Controller of Explosives
For Chief Controller of Explosives
Nagpur

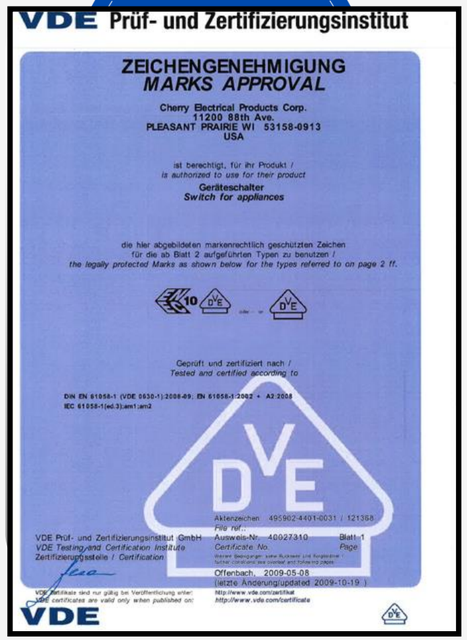
for Chief Controller of Explosives
Nagpur

Copy to :
1. JS, Chief Controller of Explosives, South Circle Office, CHENNAI

(For more information regarding status,fees and other details please visit our website http://peso.gov.in)

This is System Generated document. Signature is not required.

PESO



VDE Prüf- und Zertifizierungsinstitut

ZEICHENGENEHMIGUNG
MARKS APPROVAL

Cherry Electrical Products Corp.
11200 88th Ave.
PLEASANT PRAIRIE WI 53158-0913
USA

ist berechtigt, für ihr Produkt /
is authorized to use for their product
Geräteschalter
Switch for appliances

die hier abgebildeten markenrechtlich geschützten Zeichen für die ab Blatt 2 aufgeführten Typen zu benutzen /
the legally protected Marks as shown below for the types referred to on page 2 ff.

Geprüft und zertifiziert nach /
Tested and certified according to

DIN EN 61058-1 (VDE 0630-1):2008-09; EN 61058-1:2002 + A2:2008
IEC 61058-1(ed.3):am1:am2

Aktenzeichen 40592-4401-0031 / 121368
File No.:
Ausweis-Nr. 40027310 Blatt-1
Certificate No. *Page*
Offenbach, 2009-05-08
(letzte Änderung/*updated* 2009-10-19)

VDE Prüf- und Zertifizierungsinstitut GmbH
VDE Testing and Certification Institute
Zertifizierungsstelle / *Certification*

VDE Zertifikate sind nur gültig bei Veröffentlichung unter: http://www.vde.com/certifikat
VDE certificates are valid only when published on: http://www.vde.com/certificate

VDE

VDE

Our Journey

1911

Foundation of TVS Group by TV Sundram Iyengar with focus on transport services

1962

Formation of Sundaram Clayton limited

1994

Formation of EOU to focus on export opportunities

2000

Newer applications with Reed technology

2002

TVS Motor awarded with Deming prize (TVS Group with 5 Deming awards in total)

2005

50:50 JV between TVS & Cherry to manufacture switches

2008

Cherry acquired by ZF; JV renamed ZF Electronics TVS

2012

Identified as engineering support center for ZF global sensor requirements and custom products

2013

Business focus placed on India auto and exports

2018

TVS acquired 100% stake. Renamed as TVS Sensing Solutions Private Limited.

2019

Sales reached all-time high of USD 8.5 bn

2019

FOSS launched

2023

Electric vehicle products launch

2024

SMT Line

TVS

TVS MOBILITY

TVS SENSING SOLUTIONS



OUR PRODUCTS

EVOLUTION

Our product portfolio has evolved to focus on four core groups: Sensors, Switches, ECC and Solenoids

1995

Switches





2005

Panel Mount-global supplies (2005 to 2013)
Reed sensors







2009

Custom switches automotive/ industrial markets





2012

Specific sensors for Automotive & off-road vehicles: India / Germany

Sensors-Hall



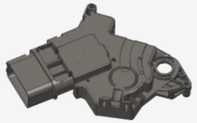
ABS sensors



E bike sensor

2015

Inroads into emerging market for ECCs & Solenoids
ECC : First player in India





Solenoids

2018

Inroads into angular sensors & actuator parts for Turbochargers (BSVI)





2019

FOPIDS introduced

FOPIDS





2020

CAS and TPMS Introduced



CAS Solenoids



TPMS

2023

Emerging Technology Focused products



DC-DC



Battery Temperature sensor



E TPS

Passenger Car

TVS MOBILITY

TVS SENSING SOLUTIONS

Turbo Charger Actuator



ABS Wheel Speed Sensor

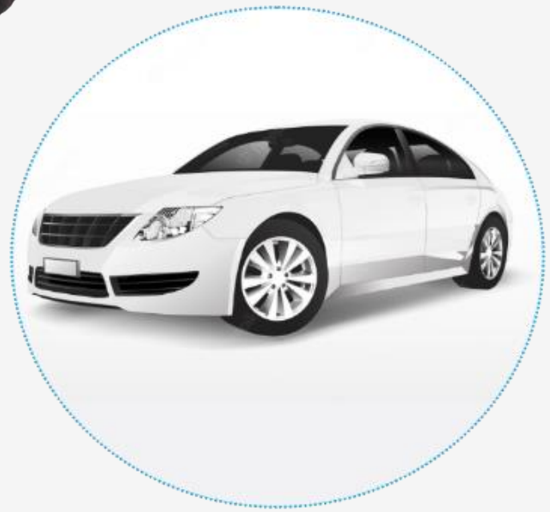




Hood Latch switch

Wiring Harness







Commutator Assembly

Tail gate Latch Assembly





Electronic component carrier



Central Locking switch





Door Latch switch

Key Reminder switch

TVS MOBILITY

Commercial & Off Road Vehicles

TVS SENSING SOLUTIONS

Linear position sensor



Speed and direction sensor



Passive Wheel Speed Sensor



Solenoid Assembly





Power Take Off switch



Cold Advance solenoid





Angular Sensor



Fuel injector Coil Assembly







Cabin tilt switch

Engine shut off Solenoid

Two Wheeler (ICE)

ABS Wheel Speed Sensor



Speed and direction sensor



Gear Position Sensor



Brake/Clutch Switch





Stop Lamp Switch





E Bike Speed Sensor



Steering Lock Control Solenoid



Plug Cap



Seat Latch Solenoid



Auto choke Solenoid



Electric Vehicles

ABS Wheel Speed Sensor



Thermistor Assembly





Acoustic Vehicle Monitoring System

Audio Visual Unit





Battery Temperature sensor



USB Charger





DC DC Converter



Battery Lock Solenoid



E Throttle position sensor



Side Stand Sensor



Tyre Pressure Monitoring Sensor

Industrial solutions



Floor level sensor



Micro switches

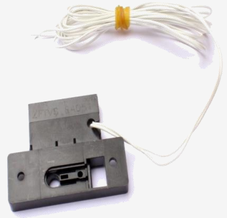


ATEX sensor



Elevator
Door Switch





NCR Reed sensor



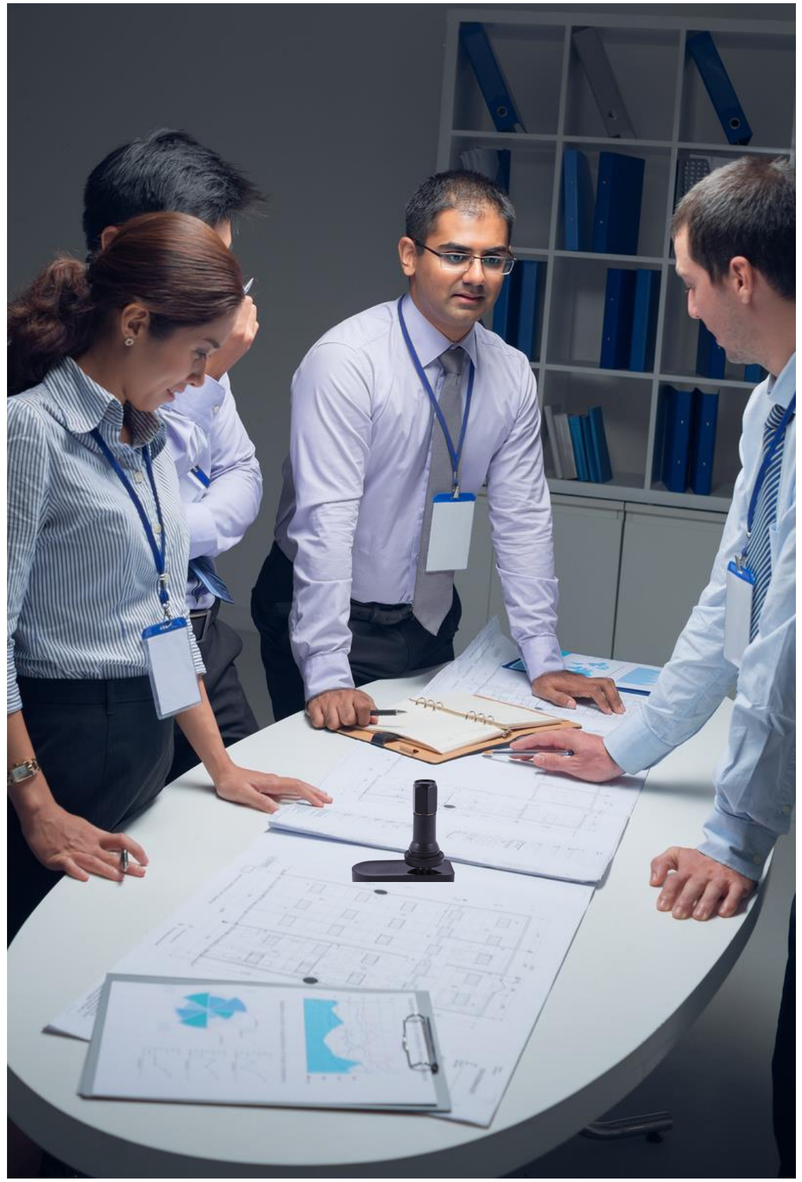
Reed sensor



Float level sensor

TVS MOBILITY

Engineering Capabilities



Application Engineering

Customization

Value Engineering

Product Knowhow

Switches, Sensors,
Solenoids and ECC

Material Knowhow

Engg Plastics, Non ferrous
metals Cu, Brass, BeCu, Silver

Design, Reengineering & Innovation capabilities

Design (No. of Seats)
Creo 2 : 4 Catia V6 : 1
Autocad : 5 Altium designer : 1

Tooling & Prototyping

A & B samples-Plastics, Metal wiring harness-Short lead-time, low investment, EDM, wire cutting

Testing & Validation

Climatic ,Salt Spray, Vibration,
Thermal Shock, Endurance, Electrical life test to UL,VDE

TVS SENSING SOLUTIONS

TVS MOBILITY

Automation Capabilities

TVS SENSING SOLUTIONS

INDUSTRY 4.0 Solutions & Services

PROCESS 360

- Process Manufacturing
- Remote Operation Centre
- Condition-based Monitoring
- Energy Optimization

OEM 360

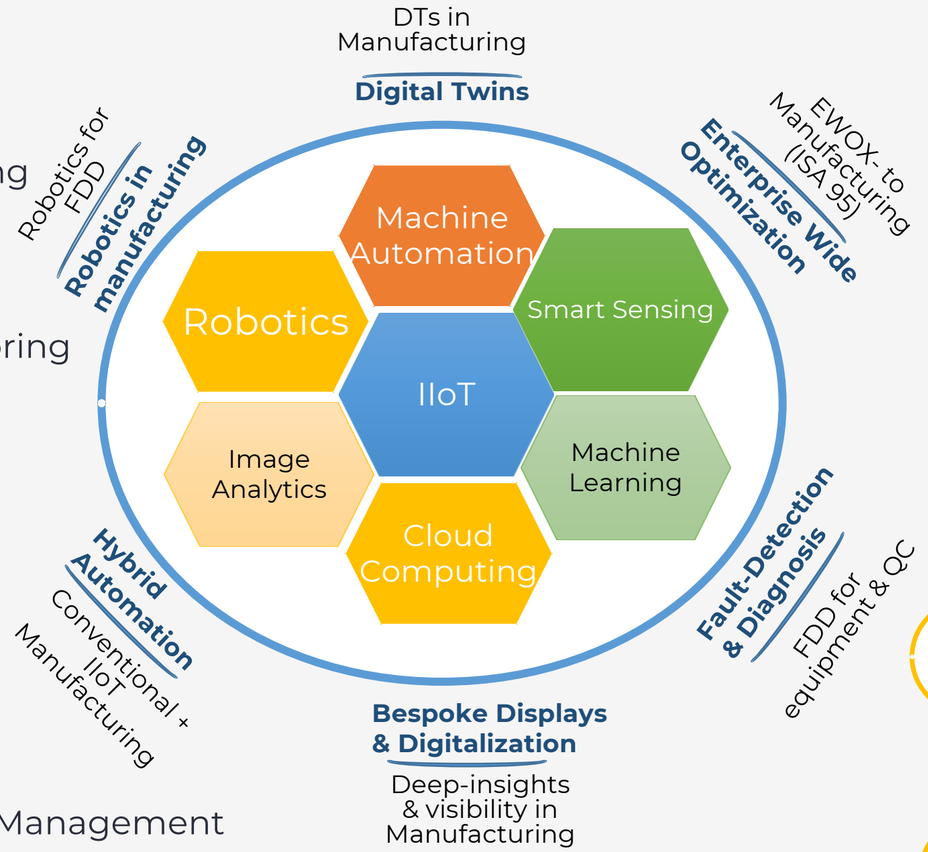
- Supplier / Customer Monitoring
- Equipment Monitoring
- User Management

DISCRETE 360

- Discrete Manufacturing
- Production Monitoring
- Quality Tracking

TVSSS EDGE

- Data Acquisition & Device Management
- Industrial Asset Connectivity
- Edge Analytics



Components: Sensors to smart sensors, IoT-based sensors, & automation components

Technology: machine learning, optimization, machine automation, analytics & cloud computing

Systems: Fault-detection & diagnosis, Enterprise wide optimization, digitalization, robotic manufacturing

Framework: Digital twin for manufacturing process, FDD systems, & other aspects



TVS MOBILITY

Manufacturing Capabilities

TVS SENSING SOLUTIONS

Switches and Sensors


Cable harness manufacturing


Crimping


Vertical Injection moulding




Horizontal and vertical Injection Moulding


Soldering robot


Coil winding


Pulse heat soldering


Dispensing equipment


Low pressure molding

SMT PCB Line : Fully automated : 46K CPH



Electronic component carriers




Stamping


Molding


Potting

Solenoids


Bobbin /Coil Molding


Resistance Welding


Crimping Core tube Assy.


Core riveting


Coil Winding


Enamel removing

QA & Reliability



Custom built EOL testers



Performance testing

Tooling


VMC


Wire EDM

Certifications

ISO 9001:2015, ISO 14001:2015 & ISO 45001:2018 certified

TVS MOBILITY

Reliability Testing Facilities

TVS SENSING SOLUTIONS



Salt Spray Chamber

Effective Volume : 250 Litres



Vibration test system

- 1000 Kgf peak
- 5 Hz to 3200 Hz
- 75g (bare table)



Thermal Shock Chamber

- Cold Zone Temp : 0°C to -70°C
- Hot Zone Temp : 50°C to 200°C



Climatic Chamber

Temp Range : -70°C to +180°Cert

TVS MOBILITY

OUR CLIENTELE

TVS SENSING SOLUTIONS

Passenger Vehicle

Tier 1

HELLA, Parker, TEL, PHA, Brakes India Private Limited, INTEVA

OEM

Mahindra, TATA, Tesla, NISSAN, Ford, HYUNDAI, FIAT, NISSAN, CITROËN, RENAULT

Two & Three Wheeler + EV

Tier 1

UNO MINDA, brembo, UCAL FUEL SYSTEMS

OEM

TVS, BAJAJ, Mahindra electric, SUZUKI, KTM, ROYAL ENFIELD, OKAYA ELECTRIC VEHICLES

Commercial / Off Road Vehicle

Tier 1

WABCO, KNORR-BREMSE, Bendix, Stanadyne, UNO MINDA

OEM

John Deere, TAFE, TATA, JCB, TOYOTA INDUSTRIAL EQUIPMENT, SML ISUZU, ZF

Industrial / Home Appliances / Elevators

GE, Whirlpool, SCHABMÜLLER, Miele, Otis A United Technologies Compar, LARSEN & TOUBRO It's all about Imagineering, NCR, KÄRCHER, KENT Mineral RO Water Purifiers, GILBARCO VEEDER-ROOT, ZF, Godrej Consumer Products Ltd., HAVELLS, ENERCON INDIA, JANATICS Pneumatic, midco, WIND WORLD Clean Energy Forever

TVS MOBILITY

THANK YOU

TVS SENSING SOLUTIONS

Contact Information

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Head of Sales & Marketing
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Email: starson.j@tvsss.co.in

www.tvsss.co.in